Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Orasanu, Judith M. Ph.D.		
Project Title:	Enhancing Team Performance for Exploration Missions		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035-1000	**Congressional District:**	14
Comments:			
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No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fischer, Ute (Georgia Institute of Technology) Kraft, Norbert (NASA ARC)		
Grant/Contract No.:	NCC 9-58-NBPF00407		
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	AIMS: Since the beginning of human space flight, incidents relating to the psychological, behavioral, and interpersonal aspects of crew performance have jeopardized crew safety and mission success. The goal of this project is to optimize team psychosocial adaptation and performance in exploration missions (Risk #24, NASA Critical Path Roadmap, 2005). We aim to enhance the ability of crews to respond adaptively to unanticipated problems, to maintain positive social and task cohesion, and to collaborate effectively under stress. The research will (a) determine the effects of task and interpersonal stressors on team performance and cohesion; (b) identify effective team interaction and stress management strategies; (c) determine the effects of individual traits and team composition on team cohesion and performance; (d) determine individual differences in response to team and task stressors; and (e) develop countermeasures to prevent deterioration of team performance and mitigate the effects of stress.		

Task Description:	APPROACH: A teamwork simulation laboratory has been created at NASA Ames Research Center to address these issues. Teams composed of 4 or 5 members engage in Antarctic or Lunar search missions over a period of 4 days (one training day, 6 simulations over 3 days). Social and task stressors are manipulated along with team gender composition. Time synchronized data (computer-based task performance, physiological measures, audio and video recordings) support analyses of team processes and outcomes. Questionnaire data (workload, team dynamics, and individual difference measures) provide essential information concerning team composition, leadership and performance both at the individual and team levels. KEY FINDINGS: During Year 2 data collection was completed for a study involving all male, all female, or mixed-gender crews (2 females and 2 males) assigned to cooperative or competitive goal conditions. Further analyses were performed on data collected in our prior study involving teams of 4 males. Findings from Year 2 address (a) effects of team composition on team performance and team climate, (b) factors that contribute to emergent leadership; (c) relations between team planning and team performance; and (d) the effect of base coordinator variables on team performance. a. Overall, gender composition did not influence task performance or team climate. However, all-female teams were more negatively affected by competitive goals under difficult task conditions and performed most poorly; gender-mixed teams performed best. Group-level personality was related to group climate, but individual-level personality was not, except in the case of the base coordinator (BC), whose Agreeableness contributed to group cohesion. Team average Agreeableness was marginally negatively related to Group Expressiveness Climate. Cooperative goals resulted in higher levels of group cohesion and marginally higher group communication efficacy than competitive goals. b. Emergent leadership (as perceived by fellow crewmembers) was expected to influence team performance. Teams were most successful when one of the searchers rather than the BC emerged as the leader. However, when the BC was viewed as the leader, higher leadership scores were associated with better team performance. Emotional stability, Disagreeableness and Extraversion characterized perceived leaders. Cognitive ability predicted perceived leadership only when correlations among different cognitive factors were taken into account. c. Task success was related to the quality of team members' planning. High-performing teams were found to cope with high task stress (time pressure, severe consequences) by explicitly planning for events and by specifying each team member's role. The BC's role was crucial in this context as he kept track of the team's progress, as well as their available time and resources. d. The base coordinator plays a special role in the simulation because he must evaluate and forward critical information concerning weather and environmental conditions, notify searchers of time-critical emergency requests, and assist in restocking supplies. The BC's personality characteristics, physiological reactivity, and communication behaviors were related to team performance: BC's levels of Extraversion, Openness, and Conscientiousness influenced team performance, but only when mediated by group cohesion and team collaboration. Lower BC reactivity to stress (lower variability in both heart rate and interbeat interval) was associated with higher team performance. BC's communication in successful teams was inclusive and agreeable, providing task-relevant information while leaving actual task coordination to the searchers. In less-successful teams, BCs micromanaged and favored some while neglecting other team members. IMPLICATIONS: These findings indicate that team factors such as group goals and strategies, team average personality, team composition, and the base coordinator's characteristics contribute more to team cohesion, climate and performance than individual crewmember personality. Emergent leadership suggests a complex relationship among the players' individual characteristics, assigned roles, perceived leadership, and team performance. Results will be used to select and compose teams to assure compatibility and optimize performance, to develop tools for self-assessment and team feedback, and to develop strategies for managing stress and to assure successful task performance. PLANS: A new study comparing the effectiveness of two approaches to team training is scheduled for the coming year: 4 hours of task management training designed to develop coordination and communication strategies responsive to shifting task demands will be compared to 20 hours of team building activities to promote mutual trust and understanding. Sleep deprivation will be included in the second phase of the study to examine the impact of training on stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Most work in government and industry is conducted in teams at least a significant portion of the time. Better understanding of factors that contribute to effective and supportive team performance for space crews will have numerous benefits, especially in high-risk industries, government, and the military. These benefits include greater efficiency, fewer errors, greater productivity, and less time lost due to interpersonal frictions or stress in the workplace. Our study will indicate which instruments are valid and useful measures of team performance and group cohesion. Understanding how to measure performance, productivity, and cohesion is the first step to being able to predict those factors. Findings on optimal crew composition will be beneficial in the design of industry work teams or any other teams in which members must work closely in a collaborative environment. In addition, our study will yield information on conflict management strategies and validated team training that could be built into education programs in colleges, professional schools, medical training, and preparation for crisis-management teams (e.g., negotiators, firefighters, police, or disaster relief teams). Countermeasures that result from the present project will have broad application to teams that operate under stressful conditions such as flight crews, air traffic controllers, medical personnel, offshore oil workers, or nuclear power operators.

Task Progress:	Three major activities were accomplished during Year Two: data collection, data analysis, and preparation for a new study. Data Collection Data collection was completed for Study 2 involving a simulated Antarctic search mission. Twenty-four 4-person crews composed of 4 males, 4 females, or 2 females and 2 males were assigned to either cooperative or competitive goal conditions. Task difficulty was varied within crews (3 difficult and 3 moderate mission scenarios over 3 days). Task performance measures were collected, in addition to physiological arousal, audio and video recordings, personality and cognitive skills, and team dynamics measures. Data Analyses In Study 2, analysis of variance, path analysis, and multilevel modeling were used to assess the following: 1. Effects of personality and gender composition on team performance and cohesion under high task and goal stress conditions 2. Impact of cognitive and personality factors on emergent leadership as perceived by other team members, and leadership impact on team success 3. Effects of the base coordinator's personal characteristics on team success 4. Validities of two measures of cognitive functioning, WinSCAT and MiniCog (compared to each other and against individual task performance) Data from our prior study (12 U.S. male teams, N = 48) were further analyzed: 1. Pre-simulation strategy planning was analyzed in relation to task performance and adaptation across the 6 scenarios as a function of prior strategy success 2. Physiological data were cleaned and reanalyzed to relate physiological responses to task difficulty, subjective workload, transient task stressors and personality 3. Task and social aspects of team members' communications as well as the base coordinator's interaction strategies were analyzed in relation to team success Preparation for New Study Preparations were completed for a new study on the effects of training and fatigue on crews' ability to cope with interpersonal stressors and to work effectively as a team. Unlike our previous studies, this experiment will involve 5-member gender-mixed teams: Two subteams of 2 people will search for water on the lunar surface (as planned by NASA), supported by a base coordinator (analogous to Mission Control). Two training and two control conditions will be compared: (1) Interpersonal Relations Training involving 20 hours of exercises, didactic instruction, and feedback; (2) Team Adaptation and Coordination Training, a 4-hour program emphasizing information sharing, planning, team monitoring and feedback; (3) a team familiarity control condition involving 22 hours of live action role play (but no training), and (4) a no-training/no-familiarity control. In the second phase of the study, team members will be deprived of sleep for 36 hours either immediately prior to beginning the first simulated Moon mission or after Day 3 to assess the interactive effect of fatigue and familiarity on team performance.
Bibliography Type:	Description: (Last Updated: 05/21/2010)
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